Exact Rational Stream-Dual Certificates for a Two-Bidder, Two-Item Auction

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Abstract

We study revenue maximization for two additive bidders and two heterogeneous items when all four values are independent and uniform on $[0, 1]$. We prove the exact rational upper bound $0.8857583025\ldots$ for all Borel-measurable, payment-integrable randomized mechanisms satisfying pointwise dominant-strategy incentive compatibility (DSIC), ex-post individual rationality, and ex-post feasibility. This improves the rigorous continuous upper certificate 0.8919 reported by Jiang, Parkes, and Wang. We also construct an explicit deterministic mechanism with exact rational revenue $0.8746049864\ldots$, guaranteeing at least 98.7408% of the unrestricted optimal revenue. The upper proof separates analytic weak duality, a rational stream-function witness, and a finite exact enclosure of its continuous objective. A divergence-free, tangent-boundary correction preserves the revenue identity; adaptive tensor-Bernstein bounds then certify the resulting polynomial envelope using directed rounding and integer accumulation. Two implementations reconstruct the upper certificate. The lower construction uses opponent-conditioned menu surcharges whose allocation-deletion property preserves simultaneous feasibility, with exact integration of the base and separate replays of the added gains. Jiang, Parkes, and Wang also report approximately 0.876 revenue for fully strategyproof GemNet; this is an external computational benchmark, not an improvement claimed for our lower mechanism. The remaining exact gap is $0.0111533161\ldots$. The stream family provides valid weak-dual witnesses without requiring a complete characterization of DSIC, and the exact optimum remains open.

1 Introduction

Optimal auction design is completely understood for a single item under the classical regularity conditions of Myerson [6]. With several items, bidder types become multidimensional and dominant-strategy incentive compatibility imposes global cross-type restrictions. Even the canonical case of two additive bidders, two items, and independent uniform values has resisted an exact characterization [4, 8].

Computational work has produced strong feasible mechanisms and rigorous upper bounds. Affine-maximizer searches provide transparent deterministic benchmarks [8], while GemNet constructs fully strategyproof menu mechanisms after a compatibility transformation [9]. Jiang, Parkes, and Wang report approximately 0.876 revenue for that GemNet benchmark on the present instance. On the dual side, the same paper develops flow-conserving DSIC kernels and a continuous lifting theorem; its strict 50×50 uniform continuous certificate is 0.8919 [4]. These are external benchmarks. The GemNet value is a reported computational revenue, whereas 0.8919 is a rigorous upper certificate. We improve the upper benchmark and provide a deterministic lower mechanism with exact rational revenue accounting.

We pursue a target-specific continuous witness rather than another grid lift. Pointwise DSIC represents each bidder's truthful utility as a convex potential [7]. A radial integration identity produces a base vector field with the required distributional source. We then add a rational polynomial curl generated by a stream function. The correction has zero divergence and zero normal boundary component, so it changes neither the conservation law nor the revenue pairing. Ex-post feasibility bounds the paired allocation pointwise by the positive maximum of the two bidders' field components.

The remaining problem is a four-dimensional continuous polynomial-envelope integral. We certify it using tensor Bernstein control coefficients, exact rational construction, directed fixed-point rounding, and a certificate-aware nonuniform dyadic traversal concentrated near unresolved winner-switching boxes. The certificate proves the upper bound $0.8857583025\ldots$. A separate implementation reconstructs the polynomial charts and repeats the complete depth-21 base traversal plus selective depth-22 and depth-23 refinement without importing the formal verifier. On the primal side, exact rational polytope and menu integration certify a layered lower mechanism with twenty common-fee rows, 41 bundle-pivot cells, and eight non-common item-containment cells. Its revenue guarantee is at least 98.7408% of the unrestricted optimum. The key structural step is deletion containment: menu perturbations can change a bidder's allocation only to a subset of its feasible predecessor allocation.

Contributions. The paper makes four contributions:

1. an explicit 32-parameter rational stream correction that yields a valid continuous DSIC weak-duality witness for the canonical two-bidder, two-item uniform instance;

2. an exact Bernstein-based integration certificate, including directed rounding, overflow bounds, deterministic transcripts, and an independent full-depth implementation replay;

3. an explicit deterministic layered taxation mechanism whose pointwise DSIC, ex-post IR, and feasibility follow from the menu and deletion arguments, with base revenue certified by one exact polytope verifier and additional revenue layers checked by separate rational replays sharing that base; and

4. an exact rational bracket of width $0.0111533161\ldots$ and a 98.7408% revenue guarantee for the explicit deterministic mechanism.

2 Relation to prior dual formulations

Rochet's cyclic-monotonicity characterization turns DSIC into convexity of truthful utility and identifies allocation vectors with subgradients [7]. Continuous duality for the one-bidder multi-item problem has been developed through transport and partial-derivative formulations [2, 3]. These results motivate integration-by-parts certificates but do not directly enforce item supply across several bidders.

Multi-bidder dual formulations also predate the present work. Finite-type flow-induced virtual values arise in unified Bayesian mechanism-design duality [1]; Zuo considers both Bayesian and dominant- strategy formulations [10]; and Beckmann-style transport gives a continuous strong-duality formulation for multi-item, multi-bidder BIC auctions in interim space [5]. The latter problem uses weaker incentive and individual-rationality requirements than the pointwise DSIC and ex-post IR requirements studied here.

Most directly, Jiang, Parkes, and Wang construct nonnegative, flow-conserving DSIC deviation kernels, derive continuous weak duality, and lift coarse uniform-grid certificates to continuous type spaces [4]. Our proof shares the same high-level weak-duality architecture: conservation eliminates payments, the dual object induces virtual coefficients, and ex-post feasibility bounds virtual surplus itemwise. The representation and certification are different. We use a signed vector field with a distributional radial source plus a divergence-free tangent stream correction, and prove its validity directly against Lipschitz convex utilities. We do not assert that every such stream field is generated by a nonnegative deviation kernel, or conversely. The contribution is an explicit rational witness for this canonical DSIC instance, an exact Bernstein certificate for its continuous objective, and a containment-based deterministic mechanism with exact revenue accounting.

3 Setting and proved bounds

Write a report profile as

$$(x, y, z, w) = (v_{11}, v_{12}, v_{21}, v_{22}) \in [0, 1]^4.$$

The two bidders are risk-neutral and additive; the two items are indivisible. The four coordinates are independent and uniform. A direct mechanism specifies jointly Borel-measurable expected allocation probabilities $x_{ij} : [0, 1]^4 \to [0, 1]$ and expected payments $p_i : [0, 1]^4 \to \mathbb{R}$, with $p_i \in L^1([0, 1]^4)$. At every report profile, itemwise feasibility requires $x_{1j} + x_{2j} \leq 1$. These marginals can be realized by assigning each item to bidder 1, bidder 2, or the seller with the corresponding probabilities. DSIC and ex-post IR are required at every type and opponent report, with utilities averaged over the mechanism's internal randomization. Let OPT be the supremum of expected revenue over this class.

Theorem 1 (Audited bracket)**.** *For the admissible class above,*

$$L = \frac{8396207869467228175603}{9600000000000000000000} \;\leq\; \mathrm{OPT} \;\leq\; U_{\mathrm{stream}},$$

where

$$U_{\mathrm{stream}} = \frac{3715139591287203}{4194304000000000} = 0.88575830251865458488464355 46875.$$

The lower endpoint is attained by the deterministic mechanism in Sections 4–5, and

$$L = 0.874604986402836268292010416 6\ldots, \qquad \frac{L}{\mathrm{OPT}} \geq \frac{L}{U_{\mathrm{stream}}} > 0.987408.$$

Thus this mechanism earns at least 98.7408% *of the unrestricted optimal revenue. The upper endpoint follows from a rational weak-dual witness and a finite exact enclosure of its objective.*

The lower bound is proved in Sections 4–6; the upper bound and a simpler radial benchmark are proved in Section 7. The stream construction supplies a valid upper certificate without assuming completeness of the chosen witness family. As an external comparison, Jiang, Parkes, and Wang report a strict continuous upper certificate 0.8919 and fully strategyproof GemNet revenue approximately 0.876 for this instance [4]. Our stream bound strictly improves their printed upper decimal by the exact rational difference

$$\frac{8919}{10000} - U_{\mathrm{stream}} = \frac{25760146312797}{4194304000000000} > 0.$$

This comparison is to their reported decimal, not a reconstruction of their underlying dual array. The 0.876 GemNet revenue is likewise a reported computational benchmark, not an exact rational certificate replayed here.

4 An exact base affine maximizer

Set

$$a = \frac{159}{250}, \qquad b = \frac{91}{100}, \qquad s = \frac{1137}{1000}.$$

For each feasible outcome A, let c_A and the affine score be as follows. The row number is also the tie-breaking priority.

id	allocation	score	c_A
0	seller keeps both	0	0
1	$1 \to 1$	$x - a$	a
2	$2 \to 1$	$y - a$	a
3	$\{1,2\} \to 1$	$x + y - b$	b
4	$1 \to 2$	$z - a$	a
5	$2 \to 2$	$w - a$	a
6	$\{1,2\} \to 2$	$z + w - b$	b
7	$1 \to 1,\ 2 \to 2$	$x + w - s$	s
8	$2 \to 1,\ 1 \to 2$	$y + z - s$	s

The base outcome $A^0(v)$ is the smallest-id maximizer of the nine scores. This defines the outcome on every equality surface.

Let

$$H_1(z,w) = \max\{0, z-a, w-a, z+w-b\}, \quad H_2(x,y) = \max\{0, x-a, y-a, x+y-b\}.$$

If $V_i(A)$ is bidder i's reported value for the items assigned in A, the base payments are

$$p_1^0 = H_1 - V_2(A^0) + c_{A^0}, \qquad p_2^0 = H_2 - V_1(A^0) + c_{A^0}.$$

Lemma 2. *The base mechanism is feasible, pointwise DSIC, and pointwise ex-post IR.*

Proof. Every listed outcome is feasible. Fix bidder 2's report. Apart from the constant H_1, bidder 1's utility from an induced outcome A, evaluated at the true type, is

$$V_1(A) + V_2(A) - c_A - H_1.$$

Truthful reporting selects a maximizer of the expression preceding H_1; no report can give greater utility. The same argument applies to bidder 2. It remains valid on ties because the selected row is still a maximizer. The outcomes excluding bidder 1 attain H_1, so the winning score is at least H_1, and bidder 1's truthful utility is nonnegative. The symmetric argument proves bidder 2's IR. $\square$

For a bidder facing opponent report $q = (q_1, q_2)$, define

$$H(q) = \max\{0, q_1 - a, q_2 - a, q_1 + q_2 - b\}.$$

The base taxation menu has prices

$$A_0(q) = H(q) + \min\{a, s - q_2\}, \quad B_0(q) = H(q) + \min\{a, s - q_1\}, \quad C_0(q) = H(q) + b \tag{1}$$

for item 1, item 2, and the bundle, respectively; opt-out costs zero. These formulas follow directly by minimizing the outcome cost net of the opponent's value among outcomes assigning the indicated bundle.

5 The entry-surcharge mechanism

Opponent-conditioned taxation menus ensure each bidder's DSIC, but their separate choices need not be simultaneously feasible. The following containment principle supplies that link.

Lemma 3 (Deletion containment)**.** *Let a deterministic predecessor mechanism have feasible allocations $S_i^0(v)$. For each bidder and opponent report, fix a finite taxation menu containing opt-out at price zero. Suppose its declared tie rule selects a utility-maximizing bundle $S_i(v) \subseteq S_i^0(v)$ at every report profile. Then the resulting mechanism is pointwise DSIC, ex-post IR, and ex-post feasible.*

Proof. Fixing the opponent report fixes all available bundles and prices. A truthful utility maximizer cannot gain by reporting another type, and opt-out guarantees nonnegative utility. Containment preserves the predecessor's disjoint item allocations. The assumptions concern the selected bundles, including ties, so all three conclusions hold pointwise. $\square$

For an opponent report $q = (q_1, q_2)$, set

$$t = \max\{q_1, q_2\}, \qquad \rho = \min\{q_1, q_2\}.$$

The active mechanism is the finite rational rule frozen in

`certificate/refined_item_containment_bundle_pivot_lower_bound/manifest.json`.

The complete evaluable tables and endpoint conventions are specified by this manifest and its hash-bound predecessor manifests. The active manifest contains the eight item-containment rows. The twenty Z/S rows and 41 bundle-pivot rows are specified, respectively, in

`certificate/piecewise_surcharge_twenty_band_lower_bound/manifest.json`
`certificate/piecewise_surcharge_bundle_pivot_lower_bound/manifest.json`.

We summarize the three-layer construction here.

First, twenty rational Z/S half-bands add a common fee $F(q)$ to all three nonempty prices in (1). If none matches, set

$$u = t + \rho, \qquad c = b - a = 137/500, \qquad d = s - a = 501/1000,$$

and first test the bundle-pivot region

$$91/100 \leq u \leq 1, \qquad c \leq \rho \leq u - d.$$

Only inside this region define $r = (\rho - c)/(u - c - d)$, whose denominator satisfies $u - c - d \geq 27/200 > 0$, and test the 41 positive cells in (u, r). Outside the region no bundle-pivot fee is applied and r need not be evaluated. Those cells also add one common fee to all nonempty prices. The Z/S rows have priority on shared boundaries. Their interiors are disjoint from the bundle cells: Z support has $u \leq b$ and S support has $\rho \leq c$, whereas a bundle-cell interior has $u > b$ and $\rho > c$.

Finally, eight rows inside the two S5 half-bands add $\delta(q)$ to the singleton corresponding to the opponent's low coordinate and to the grand bundle, leaving the other singleton unchanged. This coordinate is unique: on every item row, $t > 139/200 > 137/500 \geq \rho$. Each row is left-open and right-closed in t; at $t = 139/200$, $\delta = 0$. If maximum utility is zero, choose opt-out. At a positive tie, retain the predecessor-selected outcome whenever it remains maximizing; otherwise choose the maximizing itemwise subset with smallest id under the fixed predecessor order.

Proposition 4. *The refined surcharge mechanism is deterministic, measurable, ex-post feasible, pointwise DSIC, and pointwise ex-post IR on the entire cube, including all menu, band, cell, and item-surcharge boundaries.*

Proof. Every common fee preserves the ranking among nonempty bundles, so its selected bundle is the base bundle or opt-out. On an item row, orient the predecessor prices as

$$(A, B, C) = (d + F, t + F, t + c + F).$$

The row changes them to $(A + \delta, B, C + \delta)$, and its exact verifier checks

$$0 < \delta < A + B - C = d - c + F.$$

If the changed singleton was selected, its utility lead over the unchanged singleton is at least $A + B - C$, and hence remains positive after the change; its ranking against the bundle is unchanged. The unchanged singleton remains selected when it was selected, a bundle buyer can move only to a subset, and an opt-out cannot enter because no nonempty price falls. Thus every final bundle is an itemwise subset of its predecessor bundle, which is itself a subset of the feasible base allocation.

For each fixed opponent report the declared rule selects a utility maximizer from a taxation menu independent of the bidder's report, with opt-out at price zero. Lemma 3 therefore proves DSIC, ex-post IR, and simultaneous feasibility at each layer. The finite rational semialgebraic tables and literal endpoint priority are Borel measurable. Numerical searches selected rational rows and fees, but no discovery file is a verifier input. □

6 Exact revenue

Lemma 5 (Revenue of a four-option menu)**.** *Suppose*

$$0 \le A, B \le C \le 1, \qquad C \le A + B.$$

Let an additive type be uniform on $[0, 1]^2$*. The expected payment from the four-option menu is*

$$\begin{aligned} r(A, B, C) &= A(1 - A)(C - A) + B(1 - B)(C - B) \\ &\quad + C\left[(1 - C + B)(1 - C + A) - \frac{(A + B - C)^2}{2}\right]. \end{aligned}$$

Proof. The strict item-1 region has area $(1 - A)(C - A)$, and the strict item-2 region has area $(1 - B)(C - B)$. The bundle region starts with the rectangle $[C - B, 1] \times [C - A, 1]$ and removes a right triangle of side $A + B - C$. Multiplying the three areas by their prices proves the formula. Ties have area zero for this integral; their pointwise allocation was already fixed in Proposition 4. □

Suppose both (A, B, C) and $(A + e, B + e, C + e)$ satisfy Lemma 5. For the nonnegative common surcharges used here, the original price conditions together with $C + e \le 1$ ensure this. Direct expansion then gives the actual revenue change

$$r(A + e, B + e, C + e) - r(A, B, C) = eG - \frac{3C}{2}e^2 - \frac{1}{2}e^3, \tag{2}$$

where

$$\begin{aligned} G = {} & (C - A)(1 - 2A) + (C - B)(1 - 2B) + (1 - C + A)(1 - C + B) \\ & - \frac{(A + B - C)^2}{2} - C(A + B - C). \end{aligned}$$

For a Z-band orientation the prices and cross-section length are

$$(A, B, C) = (s - t, a, b), \qquad b - t,$$

and for an S-band orientation they are

$$(A, B, C) = (t, s - a, t + b - a), \qquad b - a.$$

Exact integration of the twenty frozen half-bands gives

$$G_Z = \frac{113291698497241117}{1000000000000000000000}, \qquad G_S = \frac{2400553481931223}{312500000000000000000},$$

and therefore the common-fee gain

$$G_Z + G_S = \frac{190109409919040253}{1000000000000000000000}.$$

On a bundle-pivot cell the prices are

$$(A, B, C) = (u - c, \rho + 227/1000, u).$$

The normalized map has Jacobian $u - 31/40$; including both orientations and both bidders gives symmetry factor four. Direct bivariate integration of the 41 positive cells gives

$$G_B = \frac{70946101529751}{10240000000000000000}.$$

The eight item-containment rows give

$$G_I = \frac{3963982653557301}{800000000000000000000}.$$

The base mechanism's exact expected revenue is

$$R_0 = \frac{26232089810531183}{30000000000000000}.$$

The exact verifier in `certificate/ama_lower_bound` reconstructs the $9 \cdot 4 \cdot 4 = 144$ affine/pivot cells using rational arithmetic, finds 39 full-dimensional cells, triangulates them into 16,336 rational four-simplices, checks total volume one, and integrates the affine payments.

The added gains are integrated layer by layer. The Z/S and bundle-pivot supports have disjoint interiors. Item-containment rows lie within the S5 support, but G_I is the change relative to the already surcharged predecessor menu, not relative to the base. Thus the revenue sum telescopes without double counting.

The active primary checker binds every predecessor by SHA-256, directly reconstructs the eight item-containment price regimes, checks their deletion slack and coverage, and integrates their exact symbolic polynomials. Its non-importing replay intersects the linear utility-comparison half-planes with $[0, 1]^2$. At rational prices, the resulting demand polygons have rational vertices, so their areas and conditional revenues are computed exactly. On each item row, the prices are affine in t and the price-regime inequalities hold throughout the interval by exact endpoint checks. Lemma 5 therefore makes each conditional revenue a polynomial of degree at most three in t. Boole's rule, which integrates polynomials of degree at most five exactly, then gives the row integral in rational arithmetic. The common-fee rows remain within that degree range after multiplication by their affine cross-section lengths. On bundle cells, the prices are affine in each of u, r; including the Jacobian gives degree at most four in u and three in r, so tensor Boole quadrature is exact. Finite differences at the sampled rational points are consistency checks, not proofs of a polynomial degree bound on the whole interval. The hash-bound sealed predecessors separately recompute the 20 common-fee rows and 41 positive bundle-pivot cells; the bundle predecessor has its own direct bivariate integration and independent scalar tensor-quadrature replay. These replays independently recompute the added gains,

Component	Revenue term	Exact integration and replay
Affine base	R_0	One rational polytope verifier
Twenty common-fee rows	$G_Z + G_S$	Polynomial integral; scalar replay
41 bundle-pivot cells	G_B	Bivariate integral; tensor replay
Eight item rows	G_I	Polynomial integral; polygon replay

Table 1: Revenue decomposition. Each added term is measured against its predecessor; all added-gain replays share the same certified base revenue.

but share the base value R_0 certified by the single polytope verifier. Table 1 summarizes the dependencies. SHA-256 bindings establish the identity of these shared inputs.

Therefore

$$R_0 + G_Z + G_S + G_B + G_I = L,$$

proving the lower half of Theorem 1.

The bundle-pivot, twenty-band, ten-band, and former two-rectangle certificates are retained as reproducible predecessors in their separate directories; none is the active lower endpoint.

7 Exact upper bounds for all randomized mechanisms

The proof has three layers: an analytic weak-duality theorem, an explicit rational witness satisfying its hypotheses, and a finite exact enclosure of the witness objective. The analytic theorem does not depend on polynomial degree, subdivision depth, or arithmetic scale.

7.1 Analytic weak duality

For own type $v_i = (x, y)$ and $r = \max(x, y)$, define the radial field

$$F^0(x, y) = \frac{3r^2 - 1}{2r^2}(x, y), \qquad F^0(0, 0) = 0.$$

Its value at the origin is immaterial to integration. Since $|F^0| = O(1/r)$, the field is integrable on the square.

Theorem 6 (Stream weak duality)**.** *For each bidder i, let G_i be a jointly C^1 vector field on $[0, 1]^4$. Suppose that, for each opponent report, its own-type divergence is zero and its normal component vanishes on every side of the own-type square. Put $F_i(v) = F^0(v_i) + G_i(v)$. Every admissible mechanism M satisfies*

$$\mathrm{Rev}(M) \le \mathcal{E}(F) := \int_{[0,1]^4} \sum_{j=1}^{2} \max\{0, F_{1j}(v), F_{2j}(v)\}\, dv. \tag{3}$$

Consequently OPT $\le \mathcal{E}(F)$.

Proof. Fix bidder i and an opponent report on a full-measure set for which the conditional expected-payment slice is integrable. Write truthful utility as $u(v)$, expected allocation as $X(v) \in [0, 1]^2$, and expected payment as $p(v)$. Pointwise DSIC gives

$$u(v) \ge u(r) + (v - r) \cdot X(r) \qquad \text{for every } v, r.$$

Thus u is convex and $X(r) \in \partial u(r)$. These inequalities and $0 \le X_j \le 1$ also make u Lipschitz. At every differentiability point, $X = \nabla u$. At nonsmooth points we retain the mechanism's given jointly measurable subgradient selection; its pointwise feasibility is part of the model, not inferred from an almost-everywhere gradient.

Joint measurability and the assumed integrability of the payment, together with boundedness of $v \cdot X$, make $u = v \cdot X - p$ measurable and integrable. Thus the conditional identities below apply on this full-measure set of opponent reports and may then be integrated by Fubini.

We retain the origin utility explicitly. The Lipschitz bound gives $|u(v) - u(0)| \leq 2$, so the opponent-dependent value $u(0)$ is integrable as well. Ex-post IR gives $u(0) \geq 0$.

Partition the square into the two radial charts

$$v = s(1,t) \quad \text{and} \quad v = s(t,1), \qquad 0 \leq s, t \leq 1,$$

each with Jacobian s. On a fixed ray set $q = (1,t)$ or $(t,1)$ and $g(s) = u(sq)$. Convexity makes g absolutely continuous and $g'(s) = q \cdot X(sq)$ almost everywhere and $g(0) = u(0)$. Since $p(sq) = sg'(s) - g(s)$,

$$\int_0^1 p(sq) s\, ds = \int_0^1 \frac{3s^2 - 1}{2} g'(s)\, ds - \frac{u(0)}{2}.$$

Combining the two charts gives the conditional revenue identity

$$\int_{[0,1]^2} p(v)\, dv = \int_{[0,1]^2} F^0(v) \cdot X(v)\, dv - u(0).$$

The correction is bounded, and $u \in W^{1,\infty}$. Weak Green integration, zero own-type divergence, and the zero normal trace therefore give

$$\int_{[0,1]^2} G_i \cdot \nabla u = 0.$$

Thus replacing F^0 by F_i preserves the conditional identity.

Apply this identity to both bidders and then integrate over opponent reports. Fubini is justified by the integrability established above. Dropping the nonpositive origin terms leaves the sum of the field-allocation pairings. For every profile and item j, feasibility gives $X_{1j} + X_{2j} \leq 1$, with both allocations nonnegative. Hence

$$\sum_i F_{ij} X_{ij} \leq \max\{0, F_{1j}, F_{2j}\}.$$

This pointwise relaxation is valid for arbitrary expected allocations and therefore for arbitrary internal randomization. Integrating proves (3); the maximizing allocation in its integrand need not itself be implementable by a DSIC mechanism. □

Boundary interpretation. The radial identity can also be checked by integration by parts. The classical divergence of F^0 is 3 in each open triangle $x > y > 0$ and $y > x > 0$, hence almost everywhere. Continuity across the diagonal away from the origin makes the interface fluxes cancel. The outer boundary flux is 2, while the flux around a puncture at the corner $(0,0)$ tends to 1, with the normal pointing into the removed square. Applying Green's formula to the two triangles outside $[0,\varepsilon]^2$ and letting $\varepsilon \downarrow 0$ gives

$$\int F^0 \cdot \nabla u = \int_{x=1} u + \int_{y=1} u - 3 \int_{[0,1]^2} u + u(0).$$

Lipschitz continuity replaces the trace at the puncture by $u(0)$ with vanishing error. This accounts for the corner term without treating a boundary corner as an interior distributional source.

7.2 An uncorrected radial benchmark

Taking $G_i = 0$ gives a useful analytic benchmark. In the radial charts, $F^0(sq) = \phi(s)q$, where $\phi(s) = (3s^2 - 1)/(2s)$.

For completeness, the remaining scalar expectation can be evaluated exactly. Under the two radial charts, a uniform type can be generated by taking S with density $2s$, an independent fair choice of which coordinate is maximal, and an independent $T \sim U[0,1]$. For one fixed item its radial coordinate therefore has distribution

$$Q \sim \frac{1}{2}\delta_1 + \frac{1}{2}U[0,1].$$

Set $Y = \max\{0, \phi(S)\}$, independently of Q. For two bidders let $a = \max(Y_1, Y_2)$ and $b = \min(Y_1, Y_2)$. Conditioning on the four atom/uniform cases for Q_1, Q_2 gives

$$\mathbb{E}_{Q_1,Q_2} \max\{Y_1Q_1, Y_2Q_2\} = \frac{3}{4}a + \frac{b^2}{6a},$$

where b^2/a is defined as zero when $a = 0$. Since the two items are symmetric,

$$U_{\mathrm{rad}} = \frac{3}{2}\mathbb{E}[a] + \frac{1}{3}\mathbb{E}\left[\frac{b^2}{a}\right]. \tag{4}$$

Put $r = 1/\sqrt{3}$. The function ϕ is increasing from zero to one on $[r, 1]$, while $Y = 0$ below r. The larger of two independent S's has density $4s^3$, hence

$$\mathbb{E}[a] = \int_r^1 4s^3\phi(s)\,ds = \frac{8}{15} + \frac{4}{45\sqrt{3}}.$$

Ordering the two S's gives the second moment as the explicit integral

$$\mathbb{E}\left[\frac{b^2}{a}\right] = 8\int_r^1 \frac{s}{\phi(s)}\left(\int_r^s t\phi(t)^2\,dt\right) ds.$$

The inner integral is

$$J(s) = \frac{9}{16}s^4 - \frac{3}{4}s^2 + \frac{1}{4}\log s + \frac{3}{16} + \frac{1}{8}\log 3,$$

so the last display equals

$$16\int_r^1 \frac{s^2 J(s)}{3s^2 - 1}\,ds.$$

Set $x = \sqrt{3}s$. The cancellation of the $\log 3$ terms in J reduces this integral to

$$\frac{1}{3\sqrt{3}}\int_1^{\sqrt{3}} x^2(x^2 - 3)\,dx + \frac{4}{3\sqrt{3}}\int_1^{\sqrt{3}} \frac{x^2 \log x}{x^2 - 1}\,dx.$$

Use $x^2/(x^2 - 1) = 1 + 1/(x^2 - 1)$. For the only non-elementary term, the substitution $y = (x-1)/(x+1)$ has upper endpoint $d = 2 - \sqrt{3}$ and gives

$$\int_1^{\sqrt{3}} \frac{\log x}{x^2 - 1}\,dx = \frac{1}{2}\int_0^d \frac{\log(1+y) - \log(1-y)}{y}\,dy = \frac{1}{2}[\mathrm{Li}_2(d) - \mathrm{Li}_2(-d)],$$

where

$$\frac{d}{dz}\,\mathrm{Li}_2(z) = -\frac{\log(1-z)}{z}$$

was used in the last equality. The remaining polynomial and $\int \log x\,dx$ terms then give

$$-\frac{26}{15} + \frac{8}{5\sqrt{3}} + \frac{4}{3}\log\sqrt{3} + \frac{2}{3\sqrt{3}}\big[\mathrm{Li}_2(d) - \mathrm{Li}_2(-d)\big].$$

Substitution in (4) gives

$$U_{\mathrm{rad}} = \frac{2}{9} + \frac{2}{3\sqrt{3}} + \frac{4}{9}\log\sqrt{3} + \frac{2}{9\sqrt{3}}\left[\mathrm{Li}_2(2-\sqrt{3}) - \mathrm{Li}_2(\sqrt{3}-2)\right].$$

This gives OPT $\le U_{\mathrm{rad}} = 0.920577474278749\ldots$. We next improve this benchmark with a rational stream correction.

7.3 The rational stream witness

We now add an opponent-dependent curl. For an exponent $e = (e_0, e_1, e_2, e_3)$, put

$$m_e = x^{e_0} y^{e_1} z^{e_2} w^{e_3}, \qquad \sigma(e) = (e_1, e_0, e_3, e_2).$$

Let E be the lexicographically smaller member of every nontrivial σ-pair of total degree at most four. Define

$$h(x,y,z,w) = \sum_{e\in E} \theta_e (m_e - m_{\sigma(e)}), \quad \psi = x(1-x)y(1-y)h, \quad G = (\partial_y \psi, -\partial_x \psi).$$

The thirty-two rational coefficients are listed below in the exact basis order used by the verifier. A convex quasi-Monte Carlo search was used only to discover them; the proof defines the field by the following rational numbers and does not trust the numerical optimization.

e	θ_e	e	θ_e
$(0,0,0,1)$	$-2995385/993184$	$(0,0,0,2)$	$2337131/676164$
$(0,0,0,3)$	$5724621/964727$	$(0,0,0,4)$	$-4591777/714341$
$(0,0,1,2)$	$-834567/846478$	$(0,0,1,3)$	$6291553/970203$
$(0,1,0,0)$	$2762675/446201$	$(0,1,0,1)$	$206964/961681$
$(0,1,0,2)$	$-6090207/979156$	$(0,1,0,3)$	$6862090/913753$
$(0,1,1,0)$	$-5206994/854575$	$(0,1,1,1)$	$515889/117356$
$(0,1,1,2)$	$-3346363/843816$	$(0,1,2,0)$	$-125263/465506$
$(0,1,2,1)$	$3005651/895053$	$(0,1,3,0)$	$-1043238/468269$
$(0,2,0,0)$	$1660919/994155$	$(0,2,0,1)$	$1276309/943222$
$(0,2,0,2)$	$-2075136/255347$	$(0,2,1,0)$	$404816/59269$
$(0,2,1,1)$	$250988/352269$	$(0,2,2,0)$	$3825425/994154$
$(0,3,0,0)$	$-5383501/354845$	$(0,3,0,1)$	$3312081/940771$
$(0,3,1,0)$	$-4885471/759831$	$(0,4,0,0)$	$7648954/974695$
$(1,1,0,1)$	$-603823/508896$	$(1,1,0,2)$	$417332/188831$
$(1,2,0,0)$	$17884893/955196$	$(1,2,0,1)$	$992961/785407$
$(1,2,1,0)$	$2761171/712822$	$(1,3,0,0)$	$-3841999/611370$

Give bidder 2 the same construction with (x,y) and (z,w) swapped, and put $F_i = F_i^0 + G_i$.

Lemma 7 (Rational witness feasibility)**.** *These fields satisfy Theorem 6. They are equivariant under bidder exchange and simultaneous item exchange.*

Proof. The corrections are polynomials. Equality of mixed partial derivatives gives $\mathrm{div}_{x,y}\, G = 0$. Moreover $G \cdot n = 0$ on each side of the own-type square: on a vertical side $\partial_y \psi = 0$, and on a horizontal side $-\partial_x \psi = 0$, because of the boundary factors in ψ. Bidder symmetry follows from the definition of G_2. Finally, $h(y,x,w,z) = -h(x,y,z,w)$, hence the same antisymmetry holds for ψ. Differentiation gives

$$G(y,x,w,z) = (G_y(x,y,z,w), G_x(x,y,z,w)),$$

where G_x, G_y denote the two components. The radial field has the same item-exchange property, proving the claim. □

7.4 Finite exact enclosure

Proposition 8 (Certified envelope)**.** *For the fields in Lemma 7,* $\mathcal{E}(F) \le U_{\text{stream}}$*.*

Computer-assisted proof. It remains to bound the integral in (3) exactly. Split each bidder square into (s, st) and (st, s). In each of the four chart pairs, multiplication by the Jacobian $s_1 s_2$ turns both item-1 competitors into exact rational polynomials in (s_1, t_1, s_2, t_2). The anti-symmetry of h under $(x, y, z, w) \mapsto (y, x, w, z)$ exchanges the two field components; invariance of the uniform distribution therefore makes the item-2 integral identical. The two charts overlap only on the diagonal and at the origin, which are null sets.

The certificate converts each polynomial and their exact difference to tensor Bernstein form. A polynomial lies in the convex hull of its Bernstein control coefficients, and its box integral is the box volume times their mean. Every exact initial control c is stored as an integer $m = \lfloor 10^9 c \rfloor$ with error radius one. A degree-n de Casteljau half-box split makes n floor averages, so the rigorous radius update is $E \leftarrow E + n$. Lower bounds for a competitor and for the exact competitor difference certify a fixed winner; otherwise the maximum upper control gives a valid supremum charge.

The tree uses the maximum-variation axis above the final four levels of a depth-21 base partition and exact one-step child-bound lookahead in those final four levels. Each unresolved depth-21 leaf is split once along the best exact child-bound axis if and only if that split saves at least one integer accumulator unit; otherwise its depth-21 charge is retained. Each unresolved child of an accepted depth-22 split is then tested independently: its best depth-23 split is accepted if and only if it saves at least one further unit, and otherwise its depth-22 charge is retained. Taking the minimum of several valid upper bounds is valid. All contributions are accumulated over the common depth-23 dyadic denominator. Each accepted split replaces its parent charge by charges on two children with disjoint interiors; a retained box is charged once. The coverage check sums the leaf volumes over all four chart pairs to $4 \cdot 2^{23} = 33554432$ common-denominator units.

The manifest-driven verifier reconstructs the polynomials and reports

$$
\begin{aligned}
\text{boxes fixed before unresolved depth-21 leaves}: &\quad 404804,\\
\text{unresolved depth-21 leaves}: &\quad 462796,\\
\text{leaves refined to depth 22}: &\quad 391618,\\
\text{depth-21 leaves retained}: &\quad 71178,\\
\text{fixed depth-22 children}: &\quad 173198,\\
\text{unresolved depth-22 children}: &\quad 610038,\\
\text{depth-22 children refined}: &\quad 609951,\\
\text{depth-22 children retained}: &\quad 87,\\
\text{fixed depth-23 children}: &\quad 249438,\\
\text{unresolved depth-23 children}: &\quad 970464,\\
\text{integer accumulator}: &\quad 3715139591287203.
\end{aligned}
$$

After the four chart pairs and the item-symmetry factor two, this gives

$$\int \sum_j \max\{0, F_{1j}, F_{2j}\} \le \frac{2 \cdot 3715139591287203}{10^9\, 2^{23}} = \boxed{\frac{3715139591287203}{4194304000000000}}.$$

□

Together with Theorem 6 and Lemma 7, this proves the upper half of Theorem 1. The verifier and its clean transcript are in the following certificate directory:

`certificate/continuous_stream_degree4_two_level_nonuniform_upper_bound/`

7.5 Reproducibility and trusted computations

The publication archive separates the proof inputs from the numerical search that found them. The trusted upper-bound directory contains the rational coefficient manifest, polynomial constructor, exact verifier, non-importing replay, and captured transcripts; the root SHA-256 manifest covers these files. The verifier checks all manifest expectations and fails closed on any mismatch. Before fixed-point conversion, every polynomial and Bernstein coefficient is constructed with `fractions.Fraction`. The largest axis degree is eight, the maximum initial absolute control is 1475346733, and the maximum observed directed-rounding radius is 181 fixed-point units. Signed array operations are checked against $2^{63}-1$; child charges and the global accumulator use arbitrary-precision Python integers.

The non-importing iterative replay independently reconstructs the basis, chart maps, polynomial arithmetic, Bernstein conversion, subdivision, winner tests, selective-refinement rule, and coverage accounting. Both implementations visit 3,738,334 retained-tree nodes and reproduce all refinement counts, with

$$\text{accumulator} = 3715139591287203, \qquad \text{coverage} = 33554432/33554432.$$

The independence is at the implementation level: both programs share the rational input manifest and the Bernstein enclosure principle. They audit the finite arithmetic; Theorem 6 supplies the analytic weak-duality argument.

The lower-bound chain reconstructs the 144 affine/pivot cells with one exact base verifier. Separate replays recompute the added gains from twenty common-fee rows, 41 positive bundle-pivot cells, and eight item-containment rows, while retaining the shared base revenue. They use scalar rational arithmetic, finite-difference consistency checks, exact Boole quadrature, and, for the final non-common perturbation, exact demand-polygon clipping. The degree bounds needed for exact quadrature follow from the analytic menu formula and the verified price regions, not from finite sampling. The supported release entry point is `python -B verification/reproduce_all.py`. It reruns all publication-facing checks, compares their structured outputs with the theorem endpoints, and verifies the archive hash manifest. The computational trusted base comprises Python's integer semantics, `fractions.Fraction`, the checked NumPy integer operations, JSON parsing, SHA-256, and file I/O. The entry point rejects optimized Python execution with an explicit runtime check and a nonzero exit status; this protects the executable assertions retained in predecessor kernels. Direct kernel invocations also require non-optimized Python. No optimizer output is read on the trusted proof path.

8 Discussion and open problem

The exact upper certificate improves the reported 0.8919 continuous benchmark, while the deterministic lower mechanism achieves at least 98.7408% of the unrestricted optimal revenue. The lower contribution is an explicit containment-based construction with exact revenue accounting; the external GemNet revenue of approximately 0.876 remains higher.

The latest upper improvement came from combining the same frozen degree-four rational witness with deeper, two-level nonuniform, certificate-aware local Bernstein refinement. The lower improvement came from adding deletion-compatible common-fee, bundle-pivot, and item-containment layers to the same exactly verified affine base. Sampled objectives were discovery evidence only. The exact residual width is

$$\begin{aligned} U_{\text{stream}} - L &= \frac{342629871077793868572569}{30720000000000000000000000} \\ &= 0.011153316115818316592633l380\ldots. \end{aligned}$$

The stream support-function bound relaxes cross-type convex-gradient compatibility: a pointwise field maximizer need not be cyclically monotone. The witness is therefore an upper certificate, not a characterization of all DSIC mechanisms. In particular, increasing polynomial degree alone does not establish convergence to OPT. A useful next step is to improve the witness jointly with its local certificate geometry, or to retain more of the convex-gradient restrictions discarded by the itemwise pointwise maximum. Closing the exact primal–dual gap, and determining whether randomization improves on the best deterministic mechanism, remain open.

Data, code, and materials availability

The project repository is publicly accessible at

`https://github.com/Baymax-ray/two-bidder-two-item-dsic-certificates`.

The accompanying release archive contains all exact certificate data, deterministic verification code, transcripts, environment information, and SHA-256 manifests needed to reproduce the stated bounds. Manuscript, certificate data, and other textual materials are released under CC BY 4.0. Verification software is separately released under the MIT License. Discovery-only experiments are segregated from the trusted proof artifacts.

AI-assisted tools declaration

This manuscript was completed with the assistance of OpenAI GPT-5.6 Sol. The tool assisted with mathematical exploration, numerical candidate generation, implementation and debugging of certificate and verification code, and drafting and revision of the manuscript. AI-generated suggestions and numerical search outputs were not treated as proofs; the stated results are supported by the analytic arguments and exact certificates described here. The author retains responsibility for the content, correctness, and final presentation of this work.

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